DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTEMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 10 / *TUTORIAL* 10

1. Gestel dat 'n ewekansige steekproef met steekproefgrootte $n_1 = 8$, steekproefgemiddelde $\bar{x}_1 = 42$ en 'n steekproefstandaardafwyking $s_1 = 4$, en 'n onafhanklike ewekansige steekproef van steekproef-grootte $n_2 = 15$, uit 'n ander populasie met 'n steekproefgemiddelde van $\bar{x}_2 = 34$ en steekproefstandaardafwyking van $s_2 = 5$, beskikbaar is. Neem aan dat die populasie variansies gelyk is.
   / *Assume that you have a random sample of size* $n_1 = 8$ *with sample mean* $\bar{x}_1 = 42$ *and a sample standard deviation of* $s_1 = 4$, *and you have an independent random sample of size* $n_2 = 15$ *from another population with a sample mean of* $\bar{x}_2 = 34$ *and a sample standard deviation of* $s_2 = 5$. *Assuming that the population variances are equal.*
   (a) Wat is die waarde van die saamgevoegde-variansie *t*-toetsstatistiek om $H_0 : \mu_1 = \mu_2$ te toets?
   / *What is the value of the pooled-variance and t-test statistic for testing* $H_0 : \mu_1 = \mu_2$?
   (b) Wat is die vryheidsgrade vir hierdie toets?
   / *What is the value of the degrees of freedom for this test*?
   (c) Gebruik 'n betekenispeil van $\alpha = 0.01$, wat is die kritieke waarde vir die eenkantige toets van die hipotese, H_0: $\mu_1 = \mu_2$ teenoor die alternatief, H_1: $\mu_1 > \mu_2$?
   / *Using a level of significance of* $\alpha = 0.01$, *what is the critical value for a one-sided test of the hypothesis* H_0: $\mu_1 = \mu_2$ *against the alternative* H_1: $\mu_1 > \mu_2$?
   (d) Wat is u beslissing?
   / *What is your decision*?
   (e) Bereken die *p*-waarde vir die toets in (c) en stel u gevolgtrekking.
   / *Calculate the p-value for the test in (c) and state your conclusion.*

2. Die bestuurder van 'n gloeilampfabriek wil bepaal of daar 'n verskil in die verwagte leeftyd van gloeilampe is wat deur twee verskillende tipes masjiene vervaardig word. Die populasie standaard-afwyking van die leeftye van gloeilampe wat deur Masjien I vervaardig word, is 110 uur en dié van Masjien II, 125 uur. 'n Ewekansige steekproef van 25 gloeilampe vanaf Masjien I lewer 'n steekproefgemiddelde leeftyd van 375 uur, en 'n soortgelyke steekproef van 25 gloeilampe van Masjien II, lewer 'n steekproefgemiddelde van 362 uur. Aanvaar dat beide die populasies normaal verdeel is.
   / *The manager at a lightbulb factory wants to determine whether there is any difference in the mean lifetime of bulbs manufactured on two different types of machines. The population standard deviation of Machine I is* 110 *hours, and that of Machine II,* 125 *hours. A random sample of* 25 *light bulbs from Machine I yields a sample mean of* 375 *hours, and a similar sample of* 25 *from Machine II yields a sample mean of* 362 *hours. Assume that both populations are normally distributed.*

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(a) Deur gebruik te maak van die 0.05 betekenispeil, is daar enige bewys van ’n verskil in die gemiddelde leeftyd van gloeilampe wat deur die twee tipes masjiene vervaardig word? Gebruik die kritieke waarde benadering.
/ *Using the* 0.05 *level of significance, is there any evidence of a difference in the mean lifetime of bulbs produced by the two types of machines*? *Use the critical value approach.*

(b) Bereken die *p*-waarde in (a) en interpreteer die betekenis.
/ *Compute the p-value in (a) and interpret its meaning.*

3. Vleis, vleisbyprodukte en ander bestanddele word saamgevoeg in ’n sekere fabriek. Nadat die bestanddele gemeng is, word dit in 250 g blikkies verpak. Beskrywende statistieke t.o.v. die gewigte van twee produksielyne, vanuit twee onafhanklike steekproewe, word in die volgende tabel gegee:
/ *Meat, meat by-products and other ingredients are mixed together at a factory. After the ingredients are thoroughly mixed, the mixture is placed in 250 g cans. Descriptive statistics concerning fill weights from two production lines, from two independent samples, are given in the following table:*

	Lyn / *Line A*	**Lyn / *Line B***
$\overline{x}$	250.140	249.916
s	0.336	0.140
n	11	16

Neem aan dat die populasie variansies gelyk is. Bepaal of daar by ’n 0.05 betekenispeil bewyse is van ’n verskil tussen die gemiddelde gewigte van blikkies wat gevul word op die twee produksielyne?
/ *Assuming that the population variances are equal. Determine whether there is any evidence, at the* 0.05 *level of significance, of a difference between the mean weights of cans filled on the two lines*?

(a) Gebruik die kritieke waarde benadering.
/ Use the critical value approach.

(b) Bereken die *p*-waarde en interpreteer die betekenis.
/ *Compute the p-value and interpret its meaning.*

4. ’n Eksperimentele ontwerp vir ’n gepaarde *t*-toets maak gebruik van ’n afgepaarde steekproef, nl. 20 pare identiese tweelinge. Wat is die vryheidsgrade vir hierdie *t*-toets?
/ *An experimental design for a paired t-test has, as a matched sample,* 20 *pairs of identical twins. How many degrees of freedom are there for this t-test*?

5. ’n Eksperimentele ontwerp vir ’n herhaalde meting *t*-toets vereis ’n meting voor en na ’n spesifieke stimulus op elk van15 persone. Wat is die vryheidsgrade vir hierdie *t*-toets?
/ *An experimental design for a repeated-measures t-test requires a measurement before and after the presentation of a stimulus on each of* 15 *subjects. How many degrees of freedom are there for this t-test*?

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6. Kan studente geld spaar deur hulle boeke by Amazon.com te koop? Om hierdie moontlikheid te ondersoek, is ’n ewekansige steekproef van 15 handboeke wat gedurende die vorige semester by die Universiteit van Stellenbosch gebruik is, geneem. Die pryse van die boeke is by ’n plaaslike boekwinkel sowel as by Amazon.com vasgestel. Toepaslike versendingsgelde sowel as belasting is ook in ag geneem. Die pryse word hieronder in Amerikaanse dollar gegee.
/ *Can students save money by buying their textbooks at Amazon.com? To investigate this possibility, a random sample of* 15 *textbooks used during the previous semester at the Stellenbosch University, was selected. The prices of these textbooks at both a local bookstore and through Amazon.com were recorded. The prices for the textbooks in US Dollar, including all relevant taxes and shipping are given below:*

Handboek / *Textbook*	Boekwinkel / *Book Store*	*Amazon.com*
Access 2000 Guidebook	52.22	57.34
HTLM 4.0 with Java Script	52.74	44.47
Designing the Physical Education Curriculum	39.04	41.48
Service Management: Operations, Strategy and IT	101.28	73.72
Real Estate Appraisal	37.45	42.04
Investments	113.41	95.38
Financial Management	109.72	119.80
Real Estate Principles	101.28	62.48
The Automobile Age	29.49	32.43
Geographic Information Systems in Ecology	70.07	74.43
Physical Geography	83.87	83.81
Understanding Contemporary Africa	23.21	26.48
Early Childhood Education	72.80	73.48
System of Transcendental Idealism (1800)	17.41	20.98
Principles and Labs for Fitness and Wellness	37.72	40.43

(a) Is daar enige bewys by ’n 1% betekenispeil dat daar ’n verskil is tussen die gemiddelde pryse van boeke by die plaaslike boekwinkel teenoor dié van Amazon.com? Gebruik die kritieke waarde benadering.
/ *At the* 1*% significance level, is there evidence of a difference between the mean price of textbooks at the local bookstore and Amazon.com? Use the critical value approach.*

(b) Watter aannames is vereis vir hierdie toets om geldig te wees?
/ *What assumption was necessary to perform this test*?

(c) Stel ’n 99% vertrouensinterval beraming op vir die gemiddelde verskil in pryse. Interpreteer die interval.
/ *Construct a* 99*% confidence interval estimate of the mean difference in price. Interpret the interval.*

(d) Vergelyk die resultate van (a) en (c).
/ *Compare the results of (a) and (c).*

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7. Neem aan dat $n_1 = 100,\ x_1 = 50,\ n_2 = 100$ en $x_2 = 30.$

/ Assume that $n_1 = 100,\ x_1 = 50,\ \ n_2 = 100,$ *and* $x_2 = 30.$

(a) Is daar by die 0.05 betekenispeil enige bewys van 'n betekenisvolle verskil tussen die twee populasie proporsies? Gebruik die kritieke waarde en *p*-waarde benaderings.
/ *At the* 0.05 *level of significance, is there evidence of a significant difference between the two population proportions? Use critical value and p-value approaches.*

(b) Konstrueer 'n 95% vertrouensinterval beraming vir die verskil tussen die twee populasie proporsies.
/ *Construct a* 95*% confidence interval estimate of the difference between the two population proportions.*

8. Neem aan dat $n_1 = 100,\ x_1 = 45,\ n_2 = 50$ en $x_2 = 25.$

/ Assume that $n_1 = 100,\ x_1 = 45,\ \ n_2 = 50,$ *and* $x_2 = 25.$

(a) Is daar by die 0.01 betekenispeil enige bewys van 'n betekenisvolle verskil tussen die twee populasie proporsies? Gebruik die kritieke waarde en *p*-waarde benaderings.
/ *At the* 0.01 *level of significance, is there evidence of a significant difference between the two population proportions? Use critical value and p-value approaches.*

(b) Konstrueer 'n 99% vertrouensinterval beraming vir die verskil tussen die twee populasie proporsies.
/ *Construct a* 99*% confidence interval estimate of the difference between the two population proportions.*

9. 'n steekproef van 500 klante is ewekansig uit 'n groot metropolitaanse area geselekteer. Die doel was om verbruikersgedrag te analiseer. Die volgende vraag is onder andere gevra: "Geniet jy dit om klere te koop?" Van die 240 mans het 136 "Ja" geantwoord. Van die 260 dames het 224 "Ja" geantwoord.
/ *A sample of* 500 *shoppers was selected in a large metropolitan area. The aim was to obtain information concerning consumer behaviour. The following question was among the questions asked: "Do you enjoy shopping for clothing?" Of* 240 *males,* 136 *answered "Yes". Of* 260 *females,* 224 *answered "Yes".*

(a) Is daar enige bewyse dat die proporsie dames wat hou van klere koop beduidend groter is as dié van mans? Gebruik die 0.01 betekenispeil.
/ *Is there evidence that the proportion of females who enjoy shopping for clothing is significantly larger than that of males? Use the* 0.01 *level of significance.*

(b) Bereken die *p*-waarde in (a) en interpreteer die betekenis.
/ *Find the p-value in (a) and interpret its meaning.*

(c) Konstrueer 'n 99% vertrouensinterval vir die verskil tussen die proporsie mans en dames wat dit geniet om klere te koop.
/ *Construct a* 99*% confidence interval estimate of the difference between the proportion of males and females who enjoy shopping for clothing.*

(d) Vergelyk die resultate verkry in (a) en (c.)
/ Compare the results obtained in (a) and (c).

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10. Die *The New York Times* het verslag gelewer oor ’n studie deur die Henry J. Kaiser Family Foundation t.o.v. die rol van media in die lewens van kinders (*Dylan Loeb McClain, “Where Is Today’s Child? Probably Watching TV”, The New York Times, December 6, 1999, C1*). In een vraag in die studie word kinders gevra of hulle elke dag ’n rekenaar gebruik. Uit ’n steekproef van 1090 kinders tussen die ouderdomme van 2 en 7 jaar het 283 elke dag ’n rekenaar gebruik. Uit ’n steekproef van 2065 tussen die ouderdomme van 8 en 18 jaar het 1053 elke dag die rekenaar gebruik.
/ *The New York Times reported a study conducted by the Henry J. Kaiser Family Foundation concerning the role of media in the lives of children (Dylan Loeb McClain, “Where Is Today’s Child? Probably Watching TV”, The New York Times, December 6, 1999, C1). In one question of the study, children were asked if they use a computer each day. From a sample of* 1090 *children between the ages of 2 and 7 years,* 283 *used a computer each day. From a sample of* 2065 *children between the ages of 8 and 18 years,* 1053 *used a computer each day.*
    (a) By die 0.05 betekenispeil, is daar enige bewys van ’n betekenisvolle verskil tussen die twee ouderdomsgroepe t.o.v. die proporsie kinders wat elke dag rekenaars gebruik?
    / *At the* 0.05 *level of significance, is there evidence of a significant difference between the two age groups in the proportion of children that use a computer each day?*
    (b) Bereken die *p*-waarde in (a) en interpreteer die betekenis.
    / *Find the p-value in (a) and interpret its meaning.*
    (c) Konstrueer ’n 95% vertrouensinterval beraming vir die verskil tussen die populasie proporsie van 2- tot 7-jaar oue en 8- tot 18-jaar oue kinders wat elke dag ’n rekenaar gebruik.
    / *Construct a* 95*% confidence interval estimate of the difference between the population proportion of 2- to 7-year-old children and 8- to 18-year-old children who use a computer each day.*

11. Bepaal die boonste kritieke waarde van *F,* in elk van die volgende eenkantige toets:
/ *Determine the upper-tail critical value of F in each of the following using one-tail test:*
    (a) $\alpha = 0.05 \quad n_1 = 16 \quad n_2 = 21$
    (b) $\alpha = 0.01 \quad n_1 = 16 \quad n_2 = 21$
    (c) $\alpha = 0.05 \quad n_1 = 10 \quad n_2 = 10$
    (d) $\alpha = 0.01 \quad n_1 = 10 \quad n_2 = 10$

12. Bepaal die boonste kritieke waarde van *F,* in elk van die volgende tweekantige toets:
/ *Determine the upper-tail critical value of F in each of the following using two-tail test:*
    (a) $\alpha = 0.10 \quad n_1 = 16 \quad n_2 = 21$
    (b) $\alpha = 0.05 \quad n_1 = 16 \quad n_2 = 21$
    (c) $\alpha = 0.02 \quad n_1 = 16 \quad n_2 = 21$
    (d) $\alpha = 0.10 \quad n_1 = 10 \quad n_2 = 10$
    (e) $\alpha = 0.05 \quad n_1 = 10 \quad n_2 = 10$
    (f) $\alpha = 0.02 \quad n_1 = 10 \quad n_2 = 10$

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13. ’n Professor in die Rekeningkunde Departement van ’n sakeskool beweer dat daar baie meer variasie is in die finale eksamenpunte van die studente wat die inleidende rekeningkunde kursus as ’n vereiste neem, teenoor die studente wat die kursus as ’n hoofvak in rekeningkunde neem. Ewekansige steekproewe van 13 nie-rekeningkunde hoofvak studente (Groep I) en 10 rekeningkunde hoofvak studente (Groep II) word uit sy groot klas geselekteer, en die volgende resultate is bereken, gebaseer op die finale eksamenpunte:
/ *A professor in the Accounting Department of a business school claims there is much more variability in the final exam scores of students taking the introductory accounting course as a requirement than for students taking the course as part of a major in accounting. Random samples of* 13 *non-accounting majors (Group I) and* 10 *accounting majors (Group II) are taken from the professor’s large class, and the following results are computed based on the final exam scores:*

$$n_1 = 13,\ s_1^2 = 210.2,\ n_2 = 10,\ s_2^2 = 36.5$$

(a) Is daar by die 0.05 betekenispeil enige bewys dat die professor se bewering waar is*?*
/ At the 0.05 *level of significance, is there evidence to support the professor’s claim*?

(b) Watter aannames omtrent die twee populasies word vereis om die gebruik van die *F*-toets te regverdig?
/ *What assumptions do you make here about the two populations in order to justify your use of the F-test*?

14. Beleggers gebruik die standaardafwyking van die maandelikse opbrengs vir ’n fonds as ’n maatstaf van die risiko betrokke by ’n fonds. ’n Fonds met ’n groter standaardafwyking word met ’n hoër risiko geassosieer as ’n fonds met ’n kleiner standaardafwyking. Die standaardafwykings en steekproefgroottes van twee fondse word in die tabel hieronder gegee. Gebruik ’n betekenispeil van 5% en toets die bewering dat die Century Equity Growth fonds ’n groter populasievariansie (m.a.w. hoër risiko) as die Fidelity Growth fonds het ($H_1: \sigma_1^2 > \sigma_2^2$). Aanvaar dat die populasies normaal verdeel en onafhanklik is.
/ *Investors use the standard deviation of the monthly return for a fund as a measure of the risk associated with the fund. A fund with a larger standard deviation is considered more risky than a fund with a lower standard deviation. The standard deviations and sample sizes for two funds are provided in the table below. Use a significance level of* 5*% and test claim that the Century Equity Growth fund has a larger population variance (i.e. is more risky) than the Fidelity Growth fund* ($H_1: \sigma_1^2 > \sigma_2^2$). *Assume that the populations are normally distributed and independent.*

Fonds / ***Fund***	**Steekproefstandaardafwyking** / ***Sample standard deviation***	**Steekproefgrootte** / ***Sample size***
Century Equity Growth (1)	18.9	15
Fidelity Growth (2)	15	20

Gebruik die kritieke waarde benadering en toon alle stappe wat verband hou met die hipotese toets.
/ *Use the critical value approach and show all the steps associated with the hypothesis test.*

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